A counterexample to complete monotonicity in Poisson approximation

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Principia Math In Progress

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Task 1. Resolve the complete-monotonicity conjecture

For a probability mass function p of mean λ, write

$$D(p) = \sum_{k\geq 0} p_k \log \frac{p_k}{e^{-\lambda}\lambda^k/k!}.$$

All logarithms are natural. Yu's Conjecture 1 [1] asserts that, for every fixed $\lambda > 0$, both $D(\mathrm{Bin}(n, \lambda/n))$, $n \geq \lambda$, and $D(\mathrm{NB}(n, n/(\lambda+n)))$, $n > 0$, are completely monotonic in n. For a sequence a_n, this means

$$(-1)^r \Delta^r a_n = \sum_{j=0}^{r} (-1)^j \binom{r}{j} a_{n+j} \geq 0 \quad (r \geq 0), \qquad \Delta a_n = a_{n+1} - a_n.$$

Theorem 1. *Let $D_n = D(\mathrm{Bin}(n, 1/n))$. Then*

$$-\frac{4003}{10^{12}} < \sum_{j=0}^{84} (-1)^j \binom{84}{j} D_{4+j} < -\frac{4002}{10^{12}} < 0. \tag{1}$$

Consequently the binomial assertion, and therefore the stated conjecture, is false.

We prove the sign by a finite rational certificate. No floating-point evaluation of the alternating sum is used in the proof. The negative-binomial assertion is not decided by this counterexample.

An exact expression for the divergence

Let $K \sim \mathrm{Bin}(n, 1/n)$, $n \geq 2$, and $Y = n - K$. Then

$$\Pr(Y = y) = \frac{w_{n,y}}{n^n}, \qquad w_{n,y} = \binom{n}{y}(n-1)^y, \qquad \sum_{y=0}^{n} w_{n,y} = n^n.$$

Dividing the binomial mass at k by the Poisson mass $e^{-1}/k!$ and taking its logarithm gives

$$1 + \log(n!) - \log((n-k)!) - n\log n + (n-k)\log(n-1).$$

Since $\mathbb{E}Y = n - 1$, averaging yields

$$D_n = 1 - n\log n + (n-1)\log(n-1) + \log(n!) - \frac{1}{n^n}\sum_{y=0}^{n} w_{n,y}\log(y!). \tag{2}$$

Here $\log(0!) = \log(1!) = 0$.

Rational bounds for all logarithms

Lemma 1. *For $1 \le x \le 2$, put $t = (x-1)/(x+1)$ and*

$$A(x) = 2\sum_{k=0}^{59} \frac{t^{2k+1}}{2k+1}, \qquad B(x) = A(x) + \frac{2t^{121}}{121(1-t^2)}.$$

Then $A(x) \le \log x \le B(x)$.

Proof. The identity $\log x = 2\operatorname{arctanh} t$ and the geometric series give $\log x = 2\sum_{k\ge 0} t^{2k+1}/(2k+1)$. For $0 \le t \le 1/3$, the omitted terms are nonnegative and their sum is at most $2t^{121}/(121(1-t^2))$. □

Set $S = 10^{50}$. For $1 \le m \le 88$, write $m = 2^e x$ with $e = \lfloor \log_2 m \rfloor$ and $1 \le x < 2$, and define the integers

$$\ell_m = \left\lfloor S\big(A(x) + eA(2)\big) \right\rfloor, \qquad u_m = \left\lceil S\big(B(x) + eB(2)\big) \right\rceil.$$

The exponent e is found by integer comparisons, without evaluating a logarithm. Thus

$$\ell_m \le S \log m \le u_m.$$

Define $F_0 = G_0 = 0$ and $F_y = \sum_{m=1}^{y} \ell_m$, $G_y = \sum_{m=1}^{y} u_m$. Equation (2) now gives

$$a_n \le SD_n \le b_n \qquad (4 \le n \le 88),$$

where the following floors and ceilings involve rational numbers only:

$$a_n = \left\lfloor S - nu_n + (n-1)\ell_{n-1} + F_n - \frac{1}{n^n}\sum_{y=0}^{n} w_{n,y} G_y \right\rfloor,$$

$$b_n = \left\lceil S - n\ell_n + (n-1)u_{n-1} + G_n - \frac{1}{n^n}\sum_{y=0}^{n} w_{n,y} F_y \right\rceil.$$

The integer sign certificate

Write $c_j = \binom{84}{j}$ and set

$$L = \sum_{\substack{0\le j\le 84\\ j \text{ even}}} c_j a_{4+j} - \sum_{\substack{0\le j\le 84\\ j \text{ odd}}} c_j b_{4+j}, \quad U = \sum_{\substack{0\le j\le 84\\ j \text{ even}}} c_j b_{4+j} - \sum_{\substack{0\le j\le 84\\ j \text{ odd}}} c_j a_{4+j}.$$

The displayed rational formulas evaluate to

$$L = -400267559715299775316466341855642397554631,$$
$$U = -400267559715296274267292737889552448468935.$$

Therefore

$$-\frac{4003}{10^{12}} < \frac{L}{S} \le \sum_{j=0}^{84} (-1)^j c_j D_{4+j} \le \frac{U}{S} < -\frac{4002}{10^{12}}.$$

The inequalities at the two ends are direct integer comparisons. All indices $4, \dots, 88$ are admissible for the fixed mean $\lambda = 1$. This proves (1).

For reproducibility, the complete calculation of L, U appears below. Python's `Fraction` class performs exact rational arithmetic; all other operations are on integers. The code implements the formulas above and requires only the standard library.

```
from fractions import Fraction as F
from math import comb

S = 10**50

def floor(x):
    return x.numerator // x.denominator

def ceil(x):
    return -floor(-x)

def series(x):
    t = (x-1)/(x+1)
    a = sum((2*t**(2*k+1)/F(2*k+1)
             for k in range(60)), F(0))
    return a, a+2*t**121/(121*(1-t*t))

c, d = series(F(2))
lo, hi = [0], [0]
for m in range(1, 89):
    e = m.bit_length()-1
    a, b = series(F(m, 2**e))
    lo.append(floor(S*(a+e*c)))
    hi.append(ceil(S*(b+e*d)))

flo, fhi = [0], [0]
for m in range(1, 89):
    flo.append(flo[-1]+lo[m])
    fhi.append(fhi[-1]+hi[m])

L = U = 0
for j in range(85):
    n = 4+j
    w = [comb(n,y)*(n-1)**y for y in range(n+1)]
    assert sum(w) == n**n
    a = (S-n*hi[n]+(n-1)*lo[n-1]+flo[n]
         - F(sum(w[y]*fhi[y] for y in range(n+1)),
             n**n))
    b = (S-n*lo[n]+(n-1)*hi[n-1]+fhi[n]
         - F(sum(w[y]*flo[y] for y in range(n+1)),
             n**n))
    a, b, c = floor(a), ceil(b), comb(84,j)
    if j % 2 == 0:
        L, U = L+c*a, U+c*b
    else:
        L, U = L-c*b, U-c*a

assert L == -400267559715299775316466341855642397554631
assert U == -400267559715296274267292737889552448468935
assert F(-4003,10**12) < F(L,S) <= F(U,S)
assert F(U,S) < F(-4002,10**12)
print(L, U)
```

Scope and checks

The separate verifier `verify_counterexample.py` saves every logarithm interval and divergence interval in `certificate.json`. It also recomputes the intervals directly from the original KL sum over K, independently of the expectation formula over Y. The scaled logarithm intervals have width at most 1; the scaled divergence intervals have width at most 349. The large cancellation in the 84th difference is accounted for by the signed interval sums, so it cannot turn a numerical rounding error into the certified sign. No minimality of the order 84 is asserted.

MathDB and the inspected public claim sources were checked on 11 September 2026. The database remained open with no attached claims, and no existing matching solution claim was located. The full dated search record is in `literature-audit.md`. In particular the later Lau–Nair–Ng paper [2] explicitly retains the fixed-mean binomial conjecture; its convexity result changes a different parameter.

Tasks 2–5

2. Determine the smallest finite-difference order at which complete monotonicity fails for some admissible pair (n, λ), and certify an example at that order.

3. Classify the fixed means $\lambda > 0$ for which $D(\mathrm{Bin}(n, \lambda/n))$ fails complete monotonicity, including whether any positive mean gives a completely monotonic sequence.

4. Determine whether eventual complete monotonicity can hold after discarding finitely many terms, and relate this to the asymptotics of high-order finite differences.

5. Prove or refute the remaining negative-binomial assertion $D(\mathrm{NB}(n, n/(\lambda+n)))$ for fixed $\lambda > 0$.

References

[1] Y. Yu, *Monotonic convergence in an information-theoretic law of small numbers*, IEEE Transactions on Information Theory **55** (2009), 5412–5422. arXiv:0810.5203v4, Conjecture 1, p. 8.

[2] C. W. Lau, C. Nair and D. Ng, *A mutual information inequality and some applications*, IEEE Transactions on Information Theory (2023). doi:10.1109/TIT.2023.3285928. Author's journal manuscript, p. 10.